

MAYANK KUMARAN

☎ 217-819-2807

✉ mayank6@illinois.edu

🌐 [LinkedIn](#)

🐙 [Github](#)

📁 [Portfolio](#)

Education

University of Illinois, Urbana-Champaign

Aug. 2022 – May 2026

Bachelor of Science in Computer Engineering, GPA: 3.77

Champaign, IL

Relevant Coursework: Data Structures; Probability and Statistics with Engineering Applications; Digital Systems Laboratory; Algorithms and Models of Computation; Numerical Methods; Artificial Intelligence; Computer Systems Engineering (OS); High Frequency Trading Technology; Applied Parallel Programming

Undergraduate Teaching Assistant: Digital Systems Laboratory, Computer Systems Engineering

Experience

Research Intern

June 2024 – August 2024

Ju Li Group - Massachusetts Institute of Technology

Cambridge, MA

- Reduced labeling costs by **200%** by implementing ALEC (Active Learning with Ensemble Clusters), a cutting-edge uncertainty sampling and k-center density clustering algorithm, to develop an **open-source** Active Learning package.
- Optimized the Frankenstein Optimizer using state-of-the-art acceleration techniques and fine-tuning hyperparameters, showcasing exceptional efficiency by reducing the average number of steps by **15.5%** against the next-best method and **92%** relative to traditional approaches like Steepest Descent in several benchmark tests.
- Co-authored a paper on **Contrastive Learning** for multi-modal crystal representations, demonstrating a **33.5%** improvement in accuracy using solely academic data sources and a **15.1%** enhancement in composition accuracy compared to industry-leading models such as SciBERT, MatSciBERT and GPT-3.5 Turbo.

Software Engineer

September 2023 – January 2024

Disruption Lab at Gies Institute

Champaign, IL

- Engineered and implemented a scalable **Github API** integration using **PyGithub** to streamline communication between front-end applications and the **Kubernetes** backend, resulting in a **60%** reduction in cloud resource allocation time.
- Designed and tailored a comprehensive solution integrating a custom GitHub API with a Slack Bot using the **Bolt API**, leading to an improvement in user interface efficiency and streamlined workflows for internal users.
- Trained and deployed an advanced chatbot leveraging OpenAI's **GPT-4** API, which enhanced user engagement by **20%** and reduced manual-handled user queries by **50%**, optimizing the overall support process.

Embedded Software Engineering Intern

May 2023 – July 2023

Ion Mobility

Singapore, SG

- Collaborated closely with hardware engineers to create and integrate the MQTT Client Shell with **microcontroller** firmware, enabling a seamless communication process and increasing overall system efficiency.
- Developed and automated robust **SSL encryption** protocols using OpenSSL with the IOT middleware, enhancing data transmission security and resulted in a **30%** reduction in vulnerabilities associated with unencrypted communication.
- Tested and validated the **MQTT** Client Shell on a custom Vehicle Control Unit (VCU), ensuring seamless communication via **USB**, and enhanced developer board bring-up times by **100%** by providing thorough documentation.

Projects

FPGA Network Card | *SystemVerilog, C*

October 2024 - December 2024

- Configured a remote development environment using **JTAG** for streamlined FPGA hardware access, improving debugging efficiency, and thoroughly documented the bring-up process for future reproducibility.
- Designing a **TCP offload engine** to decrease CPU load by **~25%**, featuring packet filtering, timestamping, and **DMA optimization** using the Xilinx MAC core and high-speed transceivers, with a focus on minimizing latency.
- Developed a **low-latency driver** with kernel bypass and a custom **TCP/IP stack**, enabling real-time packet filtering.

Custom Unix Operating System | *C, Assembly, RISC-V, GDB, QEMU*

October 2024 - December 2024

- Implemented **virtual memory** with **Sv39 paging** and **preemptive multitasking** for multi-process support.
- Integrated **filesystem abstractions**, including file and device IO, and integrated **sleep locks** for synchronization to optimise thread usage by over **100%**.
- Built a system call interface for essential operations, including process management, file handling, and device control.

Mammoth Melee | *React.js, Express.js, Node.js, AWS, Websockets, Photoshop*

January 2024 - Present

- Led a team of 5 in developing a multiplayer 2D fighter game with a **React.js** frontend and a **Node/Express** backend.
- Deployed a backend infrastructure on **AWS**, utilizing EC2 instances for hosting and **CloudFront** for global delivery.
- Implemented **WebSockets** for real-time online play, enabling smooth, low-latency multiplayer experiences.

Technical Skills

Languages: SystemVerilog, Python, C/C++, JavaScript, HTML/CSS, Swift, RISC-V Assembly

Frameworks: React.js, Next.js, Node.js, Express.js, Flask, WordPress, Firebase, CUDA

Developer Tools: Git, Docker, Vivado, Xilinx SDK, VS Code, Unreal Engine, Eclipse, Unity, Vercel, GDB, QEMU

Libraries: Pandas, NumPy, SciPy, SymPy, Matplotlib, NLTK, BeautifulSoup, PyTorch, OpenCV